

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 19- 303 - 2
LOW VOLTAGE CIRCUIT BREAKERS UP TO 6300 A	Date: 1-5-2025

EDMS 19-303-2

SPECIFICATION

FOR

LOW VOLTAGE CIRCUIT BREAKERS

UP TO 6300 A

Issue: May 2025 / Rev- 2

توجد بنود اختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.

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1. SCOPE:

This specification covers the minimum technical requirements for design, engineering, manufacturing, testing and packing of the three phase low voltage circuit breakers to be used in low voltage compartment of three phase distribution kiosk and three phase distribution cabinets in the LV distribution system.

2. APPLICABLE STANDARDS:

Unless otherwise specified in this specification should be comply with the latest edition of IEC standard and should be designed, manufactured and tested in accordance to the latest applicable IEC standards as follows:

No.	Standard	Description
1	IEC60947-1	Low Voltage Switchgear and Control gear General Rules.
2	IEC 60947-2	Low Voltage Switchgear and Control gear for Circuit Breakers
3	IEEEEC37.13	For De-rating

3. ENVIRONMENTAL CONDITION

The performance of low voltage assembly (panel) should be guaranteed for following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer:

Ambient temperature	-5°C to +45°C (50°C is an option according to --EDC requirements),
Maximum relative humidity	95 %
Altitude	Up to 1000 m above mean sea-level

4. DESIGN CRITERIA

- 1- Required rated current of circuit breaker at 40°C (Optional at 50°C according to –EDC requirements)
- 2- In case of the required rated current of circuit breaker at 50°C and the tested one at 40°C, should be apply derating guide table
- 3- Circuits Breakers described in this specification shall be suitable for operation in --EDC system parameters and service conditions.
- 4- Circuit breakers should be fixed Type, Indoor, Three-pole, for surface or panel mounting.

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- 5- Air circuit Breakers with electronic control shall be equipped with electronic solid state trip-control unit for overload and short circuit protection with sealing provision, (Optional for MCCB CBs).
- 6- The circuit breaker shall have instantaneous short circuit tripping characteristics such that the fault will trip the circuit breaker in a fraction of a second.
- 7- Air circuit breakers shall be with inverse time delay thermal and instantaneous short circuit release on all poles of the circuit breakers
- 8- The circuit breakers shall have on, off and trip position
- 9- The circuit breakers shall be hand operated through handle or as required.
- 10- The insulating materials used shall be non-flammable and non-hygroscopic.
- 11- The conductive parts of circuit breakers should be made of high grade copper silver coated.
- 12- The time-current characteristics of releases should be provided with the offer.
- 13- All circuit breakers should be fitted with test facility which allows the release test to be done; this test must be carried out with circuit breaker closed and with no current.
- 14- All circuit breakers should be suitable for isolation.
- 15- All MCCB circuit breakers shall be supplied with 3 extension copper bars (or 6 bars) of vertical length not more than 7 cm with current density not more than 1.5 A/mm² and the middle bar must be in different level than other adjacent bars.(as option)
- 16- Withdrawable air Circuit Breaker (as option)
 - a. The withdrawable mechanism shall be fitted with a reliable indicating device which indicates unambiguously the positions of the isolating contacts.
 - b. The withdrawable mechanism shall be fitted with interlocks which only permit the isolating contacts to be separated or reclosed when the main contacts of the circuit-breaker are open.
 - c. In addition, the withdrawable mechanism shall be fitted with interlocks which only permit the main contacts to be closed
 - when the isolating contacts are fully closed, or
 - when the specified isolating distance is achieved between the fixed and moving parts of the isolating contacts (disconnected position).
 - d. Should be include a mechanical position indication with an acknowledge facility.
 - e. Safety shutters should be available so that it prevents the direct contact with live parts.

5. INSPECTION AND TESTING:

- 1) --EDC Company for Electricity Distribution preserves its right to witness the routine testing of circuit breakers.

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- 2) Unless otherwise specified or approved in writing by (--EDC) all circuit breakers shall be tested in accordance with all relevant latest applicable IEC standards and test report certificate shall be provided with it. (--EDC) may carryout testing at manufacturer / supplier lab. The breaker will be assumed as rejected if any functions of the breakers are found faulty.

5.1 TYPE TEST

2) Type Test certificate according to the latest IEC 60947-2 and shall include the following type tests:

- Short -circuit breaking capacity
- Temperature rise.
- Overload performance (Circuit breakers of rated up to and including 630 A).
- Short-time withstand current
- Dielectric properties.
- Tripping limits and characteristics.
- Operational performance capability.

5.2 ROUTINE TEST

- 1) All routine tests prescribed in the relevant IEC 60947-2 standard shall be carried out by the manufacturer / supplier on his expense and under the full control and supervision of (--EDC) witness engineers.
- 2) The routine test samples will be randomly selected by (--EDC) from each offered lot of the same type and rating. The test will be done at an approved testing laboratory or manufacturer /supplier laboratory as will be mutually agreed upon.
- 3) The routine test will include the following:
 - Mechanical operation tests (No-load).
 - Verification of the Calibration of over-current releases (shall be test phase by phase).
 - Verification of the operation of under voltage and shunt releases.
 - Dielectric tests.
 - Verification of clearances.

5.3 SAMPLE (ACCEPTANCE)TEST:-

- --EDC has right to make AM routine test on random samples for each shipment of delivery
- No. of samples shall be decided by –EDC at tender phase.

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6. PACKING & SHIPPING

The packing shall be as following:

- Each MCCB/ACB and its accessories shall be separately packed as a complete unit/assembly and shall be delivered ready for services.
- Packing shall be protected against damage during shipment and handling to installation site.
- Packing openings shall be closed to prevent entry of dust, dirt and other foreign matters.
- Packing shall be marked with following:
 - Manufacturer's name
 - Country of origin
 - Weight in kilogram

7. MARKING

The following data shall be marked on the Circuit Breaker itself or on a nameplate engraved or laser printed on the circuit breaker, and located in a place such that they are visible and legible when circuit breaker is installed:

- Manufacturer name
- Country of origin
- Circuit breaker type
- Serial number
- Year of manufacture
- No of IEC standard
- Rated current (I_n) at -20°C (if more than 40°C)
- Rated voltage (V_n)
- Rated thermal (I_{th}) and magnetic (I_m) current
- Rated frequency.
- Rated service short-circuits breaking capacity (I_{cs}) at 400v.
- Rated ultimate short-circuit breaking capacity (I_{cu}) at 400v.

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8. SUBMITTALS

a. Vendor shall provide the following with the quotation:

1. Shall be fill in thoroughly the attached guarantee tables.
2. Technical catalogues containing the technical data of circuit breakers
3. Installation and maintenance instructions.
4. Copy of type test certificate for circuit breaker.
5. Submitted a valid approved letter from the Egyptian Electricity Holding Company.
6. Submit with the offer sample of each item (optional according to –EDC)

b. Vendor shall provide the following with delivering:

1. Operation (user) manual.

9. GUARANTEE

- The supplier guarantees the circuit breaker against all defects arising out of faulty design or workmanship, or of defective material for period of 18 months from date of delivery or 12 months from date of operation whichever comes first.
- Guarantee period shall be duplicated for circuit breaker which deliver for first time.
- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.

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10. MCCB Table 1

No. of poles	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
frame current at 40 °C ambient temperature, A	32	40	63	100	125	160	250	250	250	250	250	250	250	250	250	250	250
Nominal current at 40°C In, A	32	40	63	80	100/125	100/160	160	200	250	320	400	400	400	400	400	400	400
Nominal breaking at 400 V, $I_{cs} \geq (I_{cu} KA \geq I_{cs})$	9/10/16/25	9/10/16/25	9/10/16/25/35/50/65	9/10/16/25/35/50/65	10/16/25/35/50/65	10/16/25/35/50/65	25/35/50/65	25/35/50/65	25/35/50/65	25/35/50/65	25/35/50/65	25/35/50/65	25/35/50/65	25/35/50/65	25/35/50/65	25/35/50/65	25/35/50/65
Thermal release setting at 40 °C, A	fixed												(0.8-1) In				
Magnetic release setting range, A	fixed												(5-10) In (fixed / adjustable)				
Rated insulation voltage, (Ui)V	690												750				
Rated impulse withstands voltage (Uimp) KV	6												8				

Note:-

- Nominal breaking current shall be not less than breaking current according table 1&2 of EDMS 02-300
- MCCB 4 poles accepted

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11. ACB. Table 2

No. of poles	3	3	3	3	3	3	3	3	3	3	3	3
frame current at 40 °C ambient temperature, A	630	800	1000	1600	1600	1600	2000	3200	3200	4000	5000	6300
Nominal current at 40°C In, A	630	800	1000	1250	1600	2000	2500	3200	4000	5000	6300	
Nominal breaking at 400 V, Ics ≥ (Icu KA ≥ Ics)	50/65						65	65	65	65	100	
Thermal release setting at 40 °C, A	(0.4-1) In											
Magnetic release setting range, A	(2-10) In											
Rated insulation voltage, (Ui)V	≥750											
Rated impulse withstands voltage (Uimp) KV	≥8											

Note:-
- ACB 4 poles accepted

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12.OPTION LIST:

The following table should be filled/attached by ...EDC during tendering.

NO.	Option	Needed	Not Needed
1	Ambient temperature: 50°C		
2	Required rated current of CB at 50 °C		
3	Electronic solid state trip-control unit for overload and short circuit protection with sealing provision for MCCB CBs).		
4	Extension copper bars & No.		
5	Withdrawable air Circuit Breaker		
6	Submit with the offer sample of each item		
7	Nominal breaking current at 400 V, Ics should be defined according toEDC requirements		

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13. Derating Guide Table

Reference Ampere Rating at 40°C (104° F)	Ampere Rating at:		
	25°C (77° F)	50°C (122° F)	60°C (140° F)
32	35	30	27
40	44	37	34
63	69	59	55
80	88	75	70
100	110	94	87
125	137	114	100
160	176	150	140
200	220	182	160
250	275	235	220
320	350	300	275
400	440	372	340
500	550	468	435
630	695	585	545
800	880	754	704
1000	1100	900	825
1250	1375	1165	1085
1600	1760	1500	1320
2000	2200	1880	1650
3200	3520	3000	2650
4000	4400	3760	3300
5000	5500	4700	4125
6300	6930	5875	5455

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14.GUARANTEE TABLE CIRCUIT BREAKERS

- Manufacturer's name :-----
- Standard specification :-----
- C.B. Type :-----
- Rated current inside the enclosure
* At 40/50 °C :----/---- A
- Rated insulation voltage : ----- V
- Rated breaking capacity at 400 V : ----- KA
- Rated making capacity, peak : ----- KA
- Rated current of thermal release inside the enclosure (Ir):
* At 40/50 °C :----/---- A
- Thermal release setting value at 40/50°C :----/---- A
- Short circuit release setting value : ----- A
- Rated insulation voltage,(Ui)V : ----- V
- One-minute power frequency withstand voltage : ----- V
- Material of main contacts :-----
- Material of arcing contacts :-----
- Type of arc control device :-----
- Material of arc control device :-----

I / We guarantee the technical data given above for the offered circuit breakers.

Signature: -----

Date: -----